

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2013-14

Zoology

Paper: ZOB-401 — Fundamental Endocrinology and Developmental Biology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Fundamental Endocrinology) (Marks: 35)

Question 1. Answer the following parts:

- () State whether the following statements are True or False. Justify your answer in 2-3 sentences: [2 × 2 = 4]
- (i) Insulin is a hyperglycemic hormone.
 - (ii) Absence of thyroid hormone causes retardation of general body growth.
- () Distinguish between the following: [2 × 2 = 4]
- (i) Neurohormones and Neurotransmitters
 - (ii) Paracrine and Autocrine modes of hormone delivery.
- () Define the following: [3 × 1 = 3]
- (i) Trophic hormones
 - (ii) Cytokines
 - (iii) Pheromones.

Question 2. Draw a well-labelled diagram of histology of ovary. Mention the chemical nature and biological actions of each hormone produced by the ovary. [4 + 2 + 6 = 12]

Question 3. Draw a labelled diagram of pituitary gland. Describe the various parts of this gland and biological actions of the glycoprotein hormones. [4 + 2 + 6 = 12]

Question 4. Write short notes on the following: [3 × 4 = 12]

- () Juvenile hormone
- () Testosterone
- () Glucocorticoid.

SECTION – B
(Developmental Biology) (Marks: 35)

Question 1. Answer the following parts:

- () Define the following: [5 × 1 = 5]
- (i) Differentiation
 - (ii) Epiboly
 - (iii) Amnion
 - (iv) Competence
 - (v) Grey crescent.
- () State whether the following statements are True or False. Justify your answer in 1–2 sentences: [3 × 2 = 6]
- (i) The majority of oocytes within the adult human ovary are maintained in the zygotene stage of meiotic prophase, often referred to as the dictyate state.
 - (ii) Numerous cortical granules are found within the central cytoplasm of the ovum.
 - (iii) Superficial cleavage is found in most of the insects.

Question 2. Answer the following parts:

- () Define the role of organizer in amphibian embryo development. [6]
- () Describe the phenomenon of embryonic induction illustrating Spemann and Mangold's experiments. [6]

Question 3. Answer the following parts:

- () Describe the fate map of chick embryo with the help of a figure. [6]
- () Describe amphibian regeneration in detail. [6]

Question 4. Answer the following parts:

- () Describe the process of external fertilization in sea urchin. [6]
- () Write about the slow block and fast block to polyspermy. [6]

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